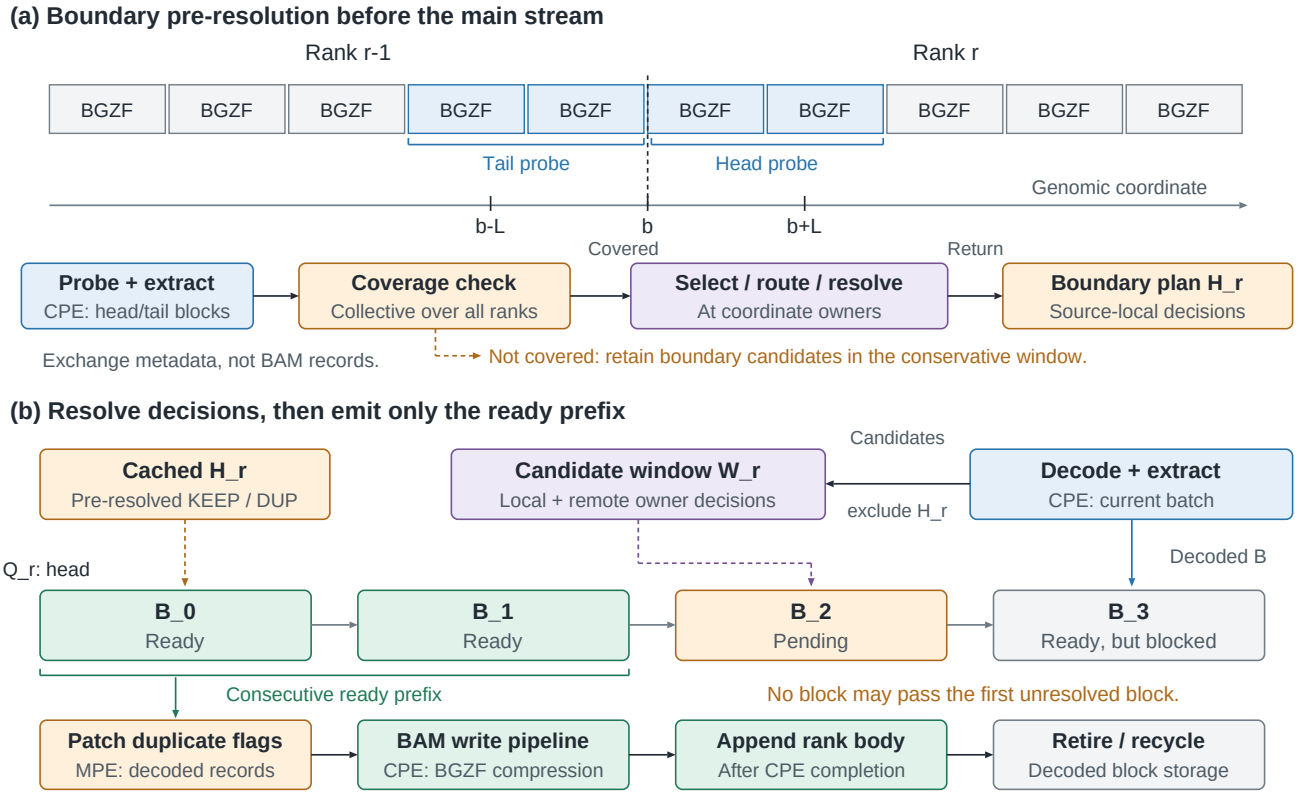
(a) Boundary pre-resolution before the main stream
Rank r-1
Rank r
BGZF
BGZF
BGZF
BGZF
BGZF
BGZF
BGZF
BGZF
BGZF
BGZF
Tail probe
Head probe
Genomic coordinate
b-L
b
b+L
Covered
Return
Probe + extract
CPE: head/tail blocks
Coverage check
Collective over all ranks
Select / route / resolve
At coordinate owners
Boundary plan H_r
Source-local decisions
Exchange metadata, not BAM records.
Not covered: retain boundary candidates in the conservative window.
(b) Resolve decisions, then emit only the ready prefix
Candidates
Cached H_r
Pre-resolved KEEP / DUP
Candidate window W_r
Local + remote owner decisions
exclude H_r
Decode + extract
CPE: current batch
Decoded B
Q_r: head
B_0
Ready
B_1
Ready
B_2
Pending
B_3
Ready, but blocked
Consecutive ready prefix
No block may pass the first unresolved block.
Patch duplicate flags
MPE: decoded records
BAM write pipeline
CPE: BGZF compression
Append rank body
After CPE completion
Retire / recycle
Decoded block storage